

## Question

The following reaction conditions are present:

1. Cu precipitates Bi from  $\text{Bi}(\text{OH})_3$  in the presence of NaCN.
2. The reaction of a methanol solution of Karl Fischer reagent ( $\text{I}_2 : \text{SO}_2 : \text{pyridine} = 1 : 3 : 10$ ) with sodium hydroxide solution yields a hydrogen sulfate ester (ignoring the formation of salts of hydrogen sulfate ester).
3.  $\text{NaBiO}_3$  oxidizes Mn(II) under acidic conditions to produce a purple-red solution.
4.  $\text{Hg}_2(\text{NO}_3)_2$  undergoes ammonolysis and disproportionation in ammonia water, and the product contains a four-membered ring cation.
5.  $\text{O}_2\text{F}_2$  reacts with  $\text{PtF}_5$  to produce an ionic compound and release gas.

Regarding the balanced and simplified chemical or ionic equations for the above reactions, which of the following statements is correct:

1. In reaction 1, Cu is oxidized to +2 valence.
2. In reaction 2, pyridine does not participate in the construction of the reaction equation.
3. After balancing the ionic equation of reaction 3, the sum of the coefficients on the left side is smaller than that on the right side.
4. The four-membered ring cation of reaction 4 is composed of four elements.
5. In reaction 5, more than 1 type of gas is produced.

A. 1, 2

B. 1, 4

C. 1, 5

**D.** 2, 3

**E.** 2, 4

**F.** 2, 5

**G.** 4, 5

## Answer

Correct Answer: **E**

## Detailed Explanation

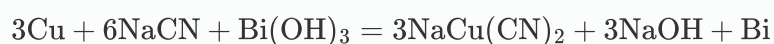
The following reactions occur:



Therefore, in reaction 1, Cu is oxidized to the +1 oxidation state. Statement 1 is incorrect.

### CHECKPOINT

2 PTS



### CHECKPOINT

1 PTS

In reaction 1, Cu is oxidized to the +1 oxidation state.



Therefore, in reaction 2, pyridine does not participate in the construction of the reaction equation. Statement 2 is correct.

## CHECKPOINT

2 PTS



## CHECKPOINT

1 PTS

In reaction 2, pyridine does not participate in the construction of the reaction equation.



Therefore, after balancing the ionic equation of reaction 3, the sum of the coefficients on the left side is greater than that on the right side. Statement 3 is incorrect.

## CHECKPOINT

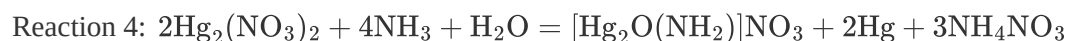
2 PTS



## CHECKPOINT

1 PTS

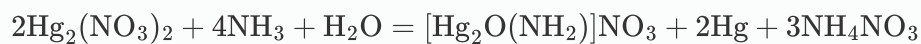
After balancing the ionic equation of reaction 3, the sum of the coefficients on the left side is greater than that on the right side



Therefore, the quaternary ring cation of reaction 4 is composed of four elements. Statement 4 is correct.

CHECKPOINT

2 PTS



CHECKPOINT

1 PTS

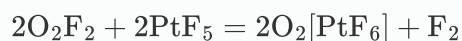
The quaternary ring cation of reaction 4 is composed of four elements



Therefore, only 1 kind of gas is produced in reaction 5. Statement 5 is incorrect.

CHECKPOINT

2 PTS



CHECKPOINT

1 PTS

Only 1 kind of gas is produced in reaction 5

Therefore, statements 2 and 4 are correct. Select option E.